



NEET 2.0 2027

Zoology

Lecture - 09

Structural Organisation in Animals

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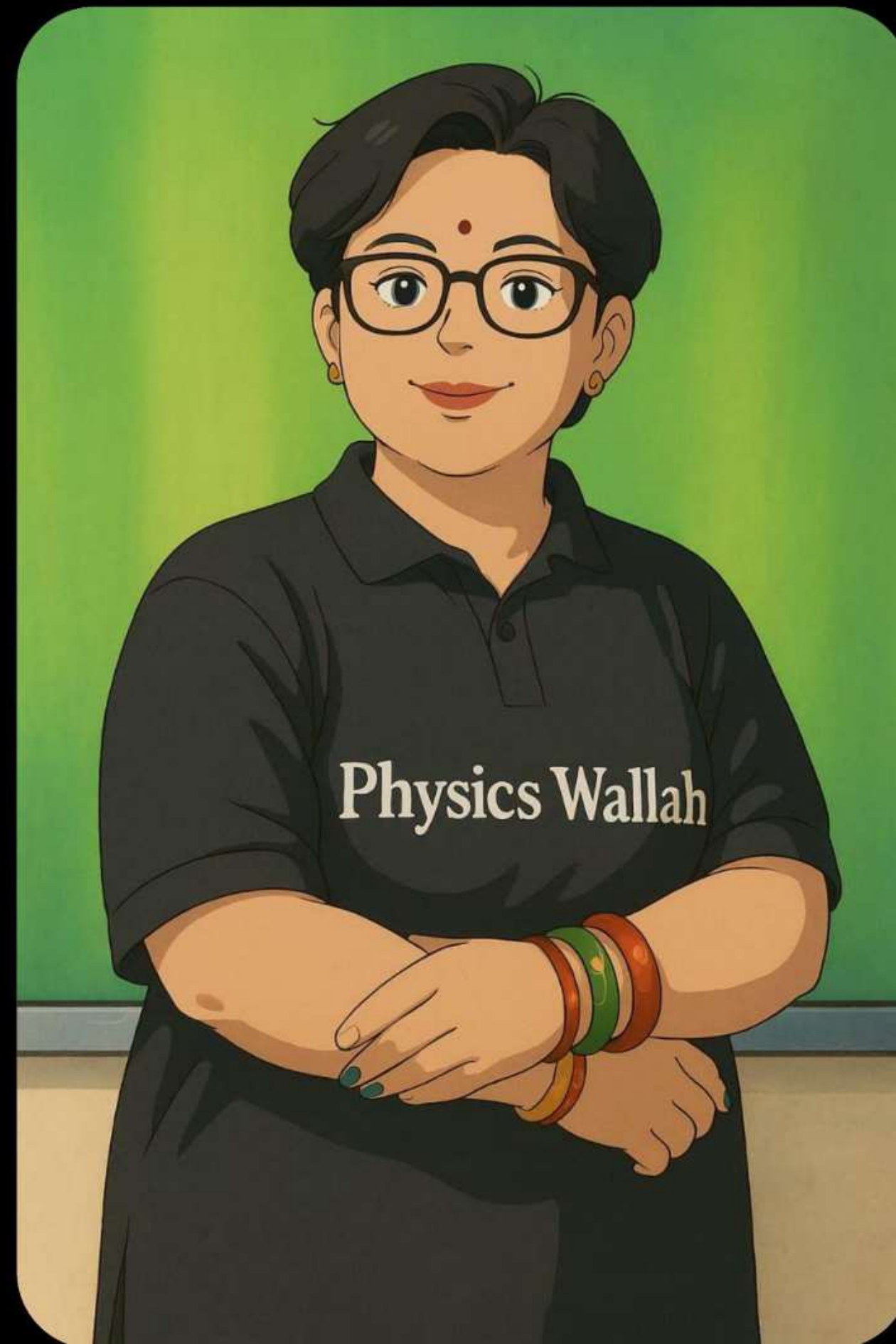


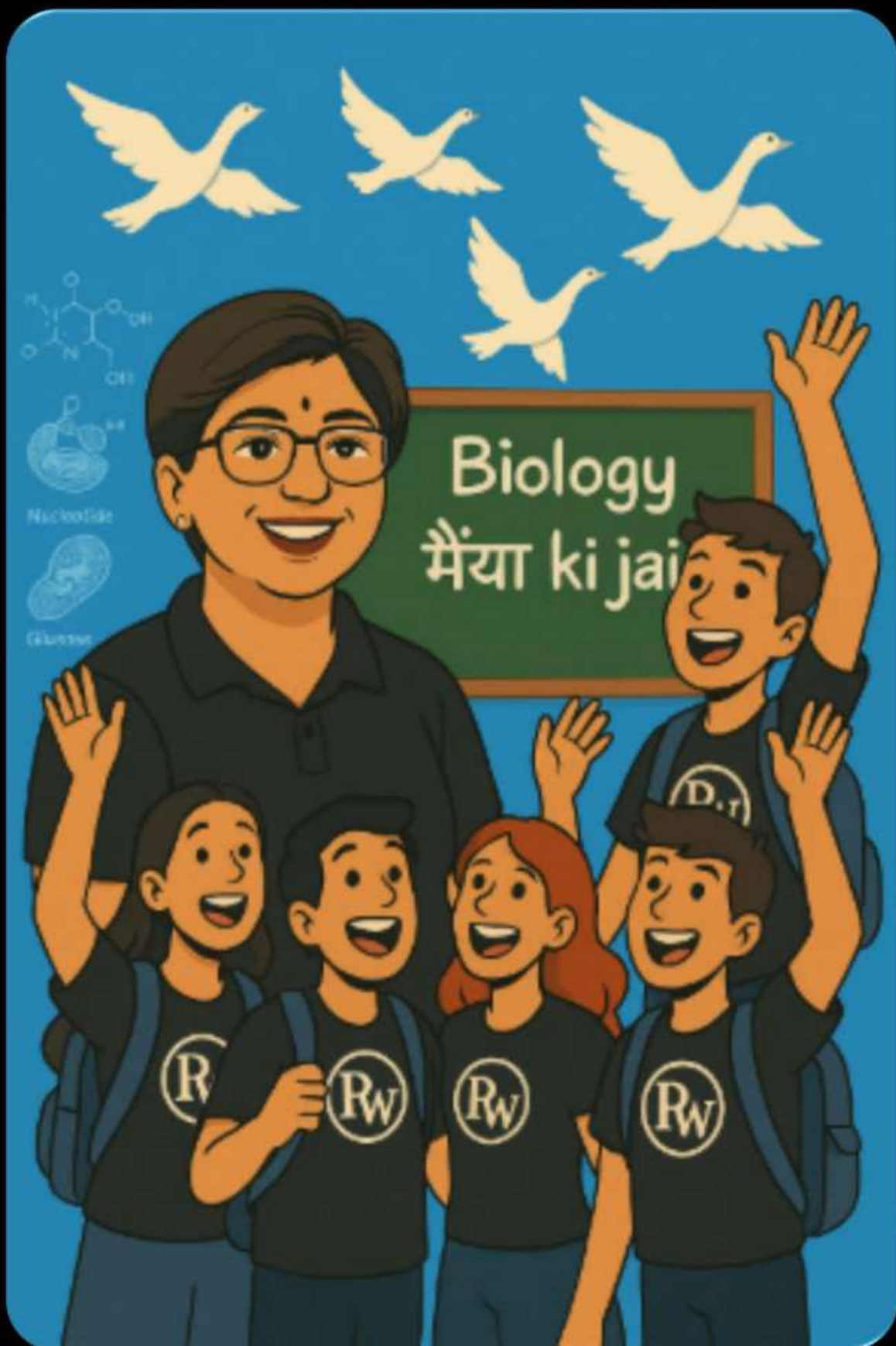


Topics to be covered

1

Frog – Morphology, Digestive, Respiratory system







@BIOLOGYBYAK
ANKSHAMAM

DA



@AAMAM_BIO

- Camouflage — Ability to change color of body,
when in grass (green) and on land (brown).
— To hide from enemies
— This protective coloration, called 'mimicry'
- Frog's Skin — Smooth and Slippery due to presence of Mucous ^(AR)
— Skin is maintained always in moist condition
- They never drink water; absorb it through skin
- Dorsal / Upper Side — Olive Green with dark irregular spots
Ventral / lower side — Pale yellow (Uniform)



Pizza

• Exhibit Sexual Dimorphism
Show 2 Morphology/External appearance

i.e. Male frog can be distinguished from Female Frog

a) Vocal Sacs ⊕
 (Croaking-Sound producing)

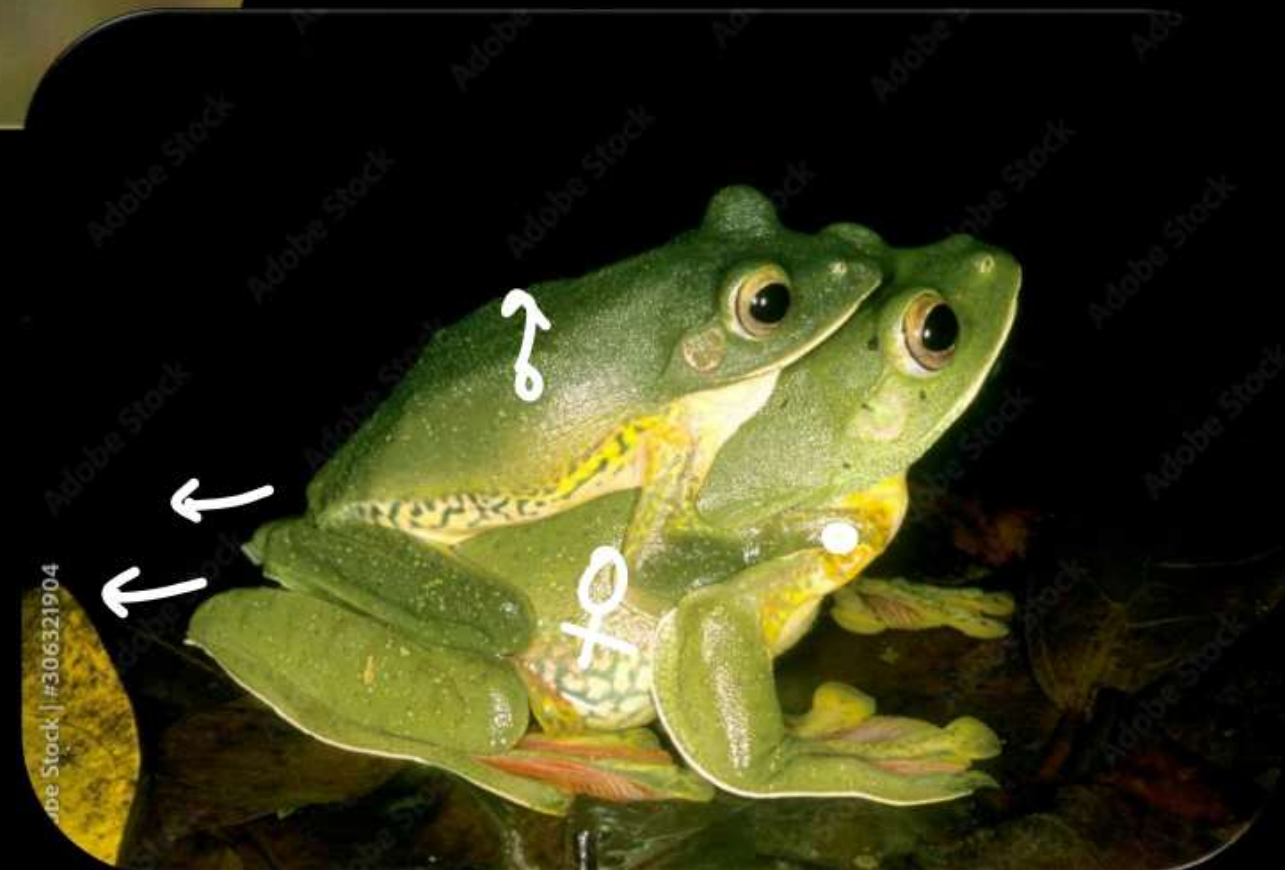
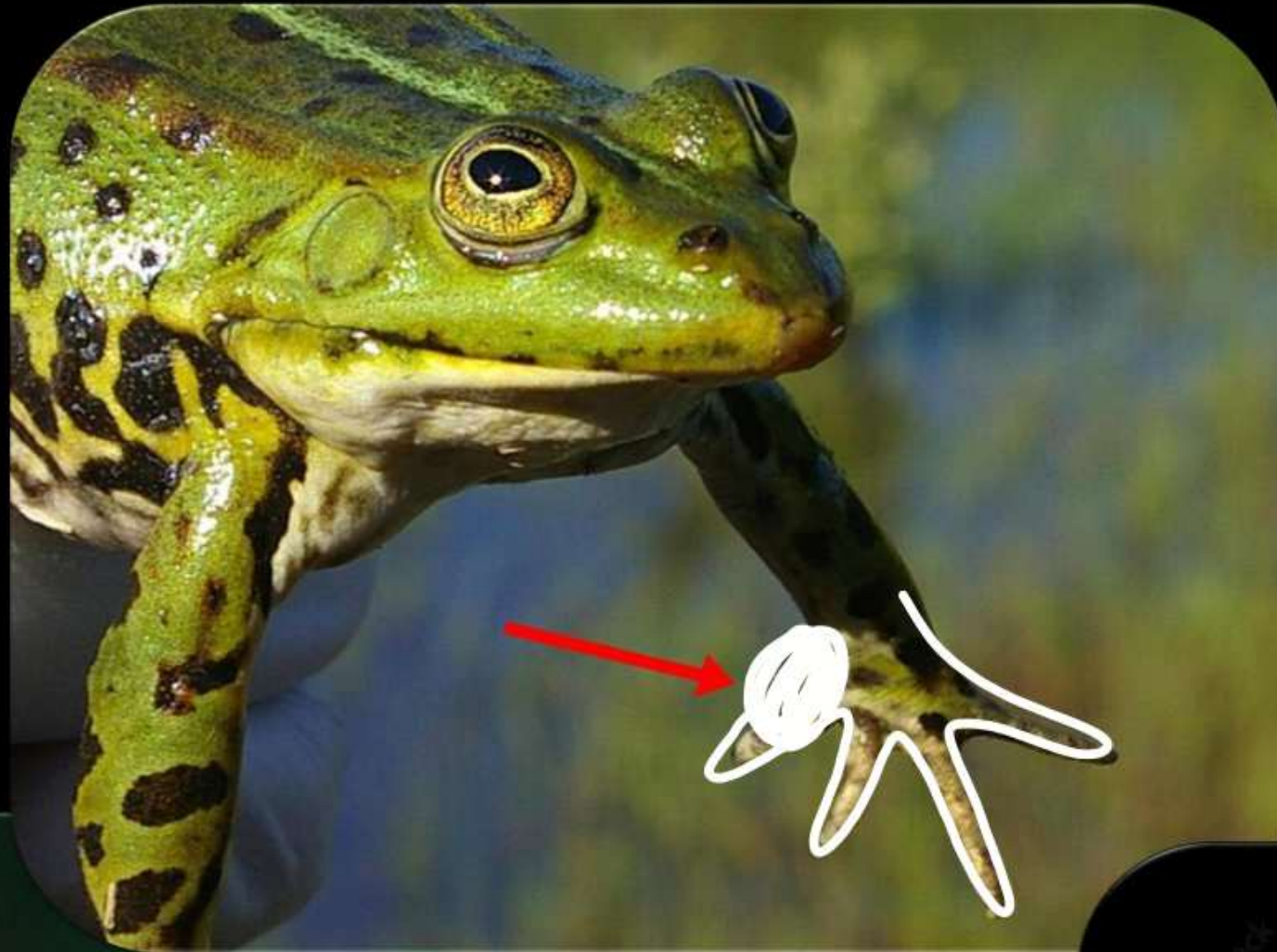
⊖

b) Copulatory Pads ⊕

⊖

2025
 PYQ On First digit of Forelimbs





* Anatomy of Frog :- (Internal Structure)

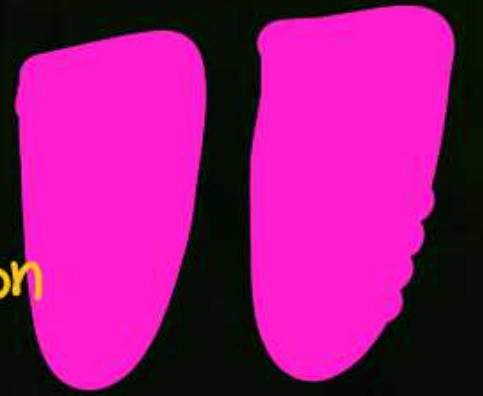


* RESPIRATORY SYSTEM : Respiration on land and in water by 2 different mechanisms

A) In Water → Air exchange ($O_2 \rightleftharpoons CO_2$) through Moist Skin
Hibernation = Cutaneous respiration
Aestivation → Dissolved O_2 in water exchanged through moist skin by Diffusion

B) On Land →
1) Moist Skin (Cutaneous)
2) Lungs (= Pulmonary resp.)
3) Buccal cavity / Bucco-pharyngeal

- LUNGS — 1 pair, Elongated, Sac-like, Pink-Colored
— Present in Upper part of Trunk region



- Atmospheric Air → NOSTRILS → Buccal Cavity — Air is pushed into — LUNGS
[• Cannot eat & breathe at same time]
[• Positive Pressure ventilation]

* DIGESTIVE SYSTEM In Frog :-



I

ALIMENTARY CANAL

- From Mouth to Cloaca
- Frogs are CARNIVORES (Meat-Eaters)

AR

That's why, Alimentary Canal Shorter

AR

length of Intestine Reduced

II

DIGESTIVE GLANDS

- a) LIVER : secrete BILE
Emulsification of Fat

large fat globule

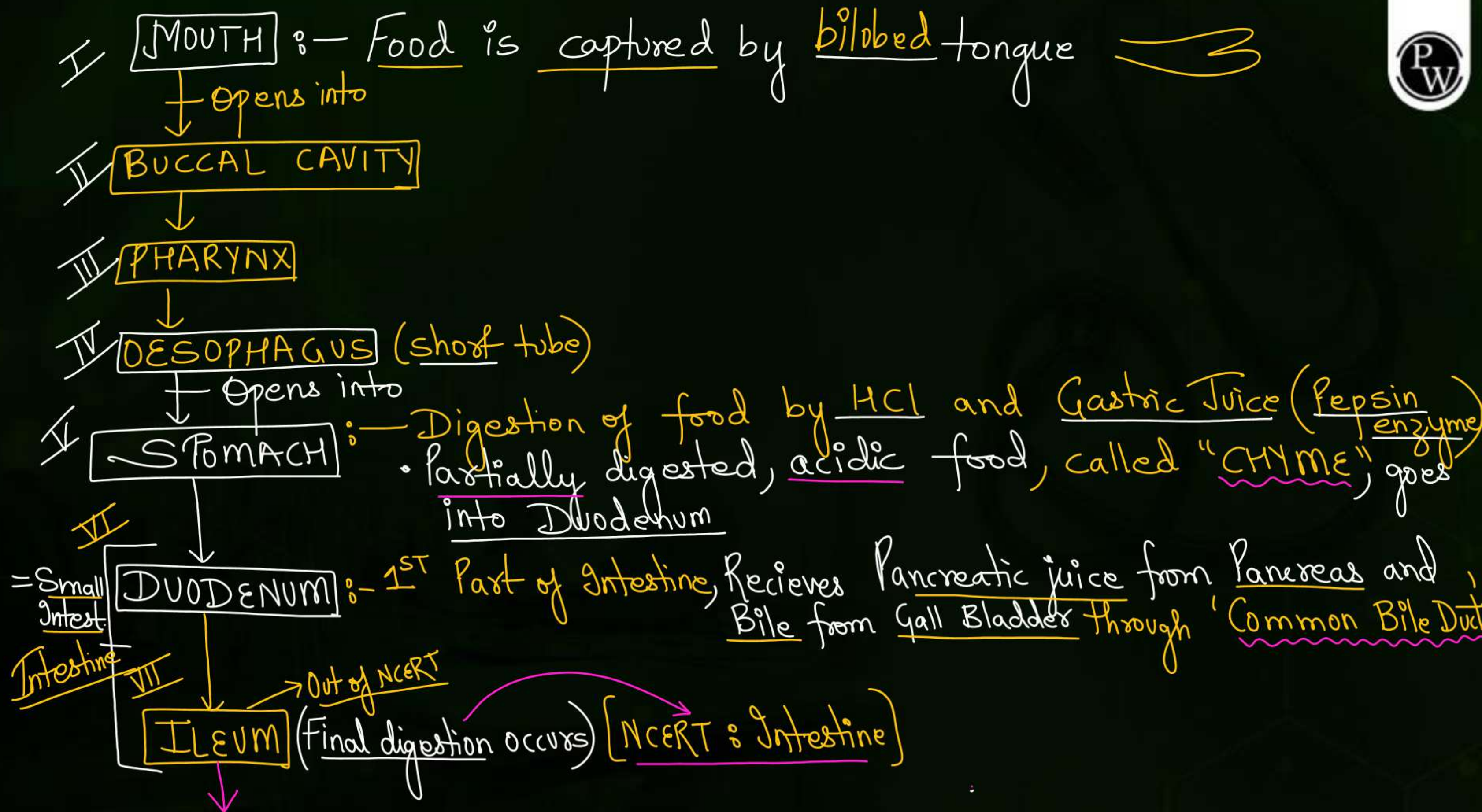
Smaller fat globules
(↑ Surface area)

- b) Gall Bladder : Store Bile

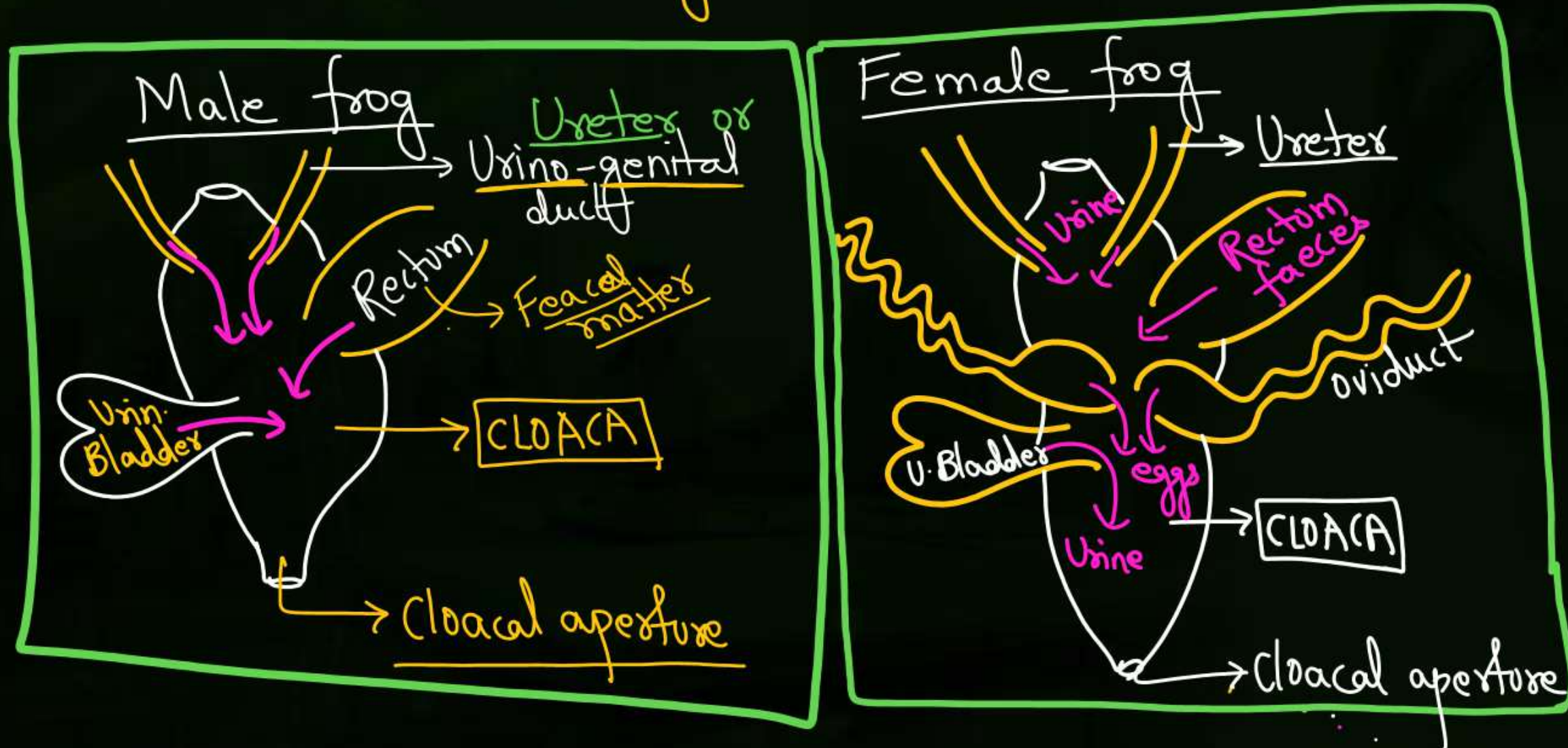
- c) Pancreas : secrete Pancreatic Juice

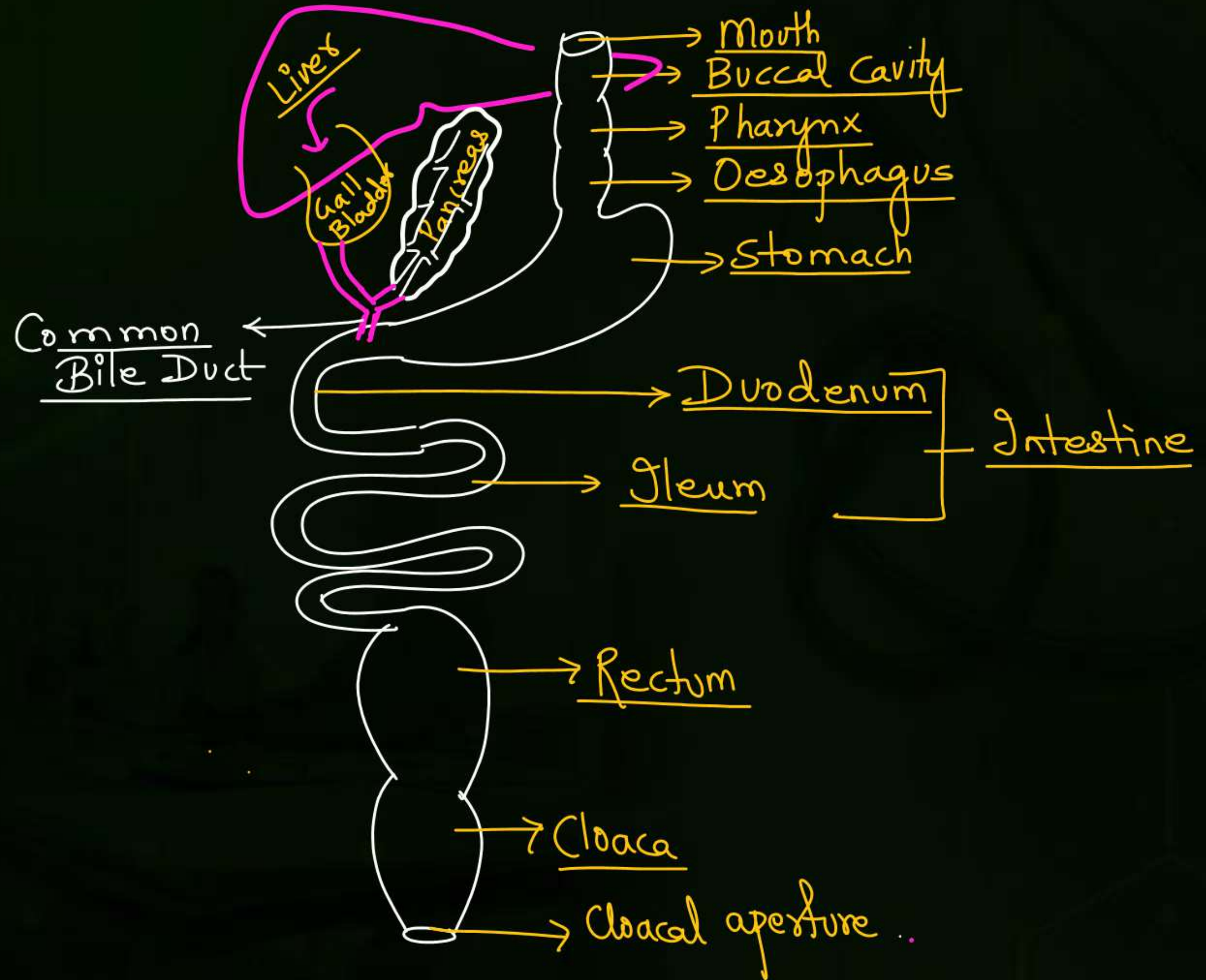
Digestive Enzymes Present

Digest Carbohydrates, Proteins
[TRYPSIN]

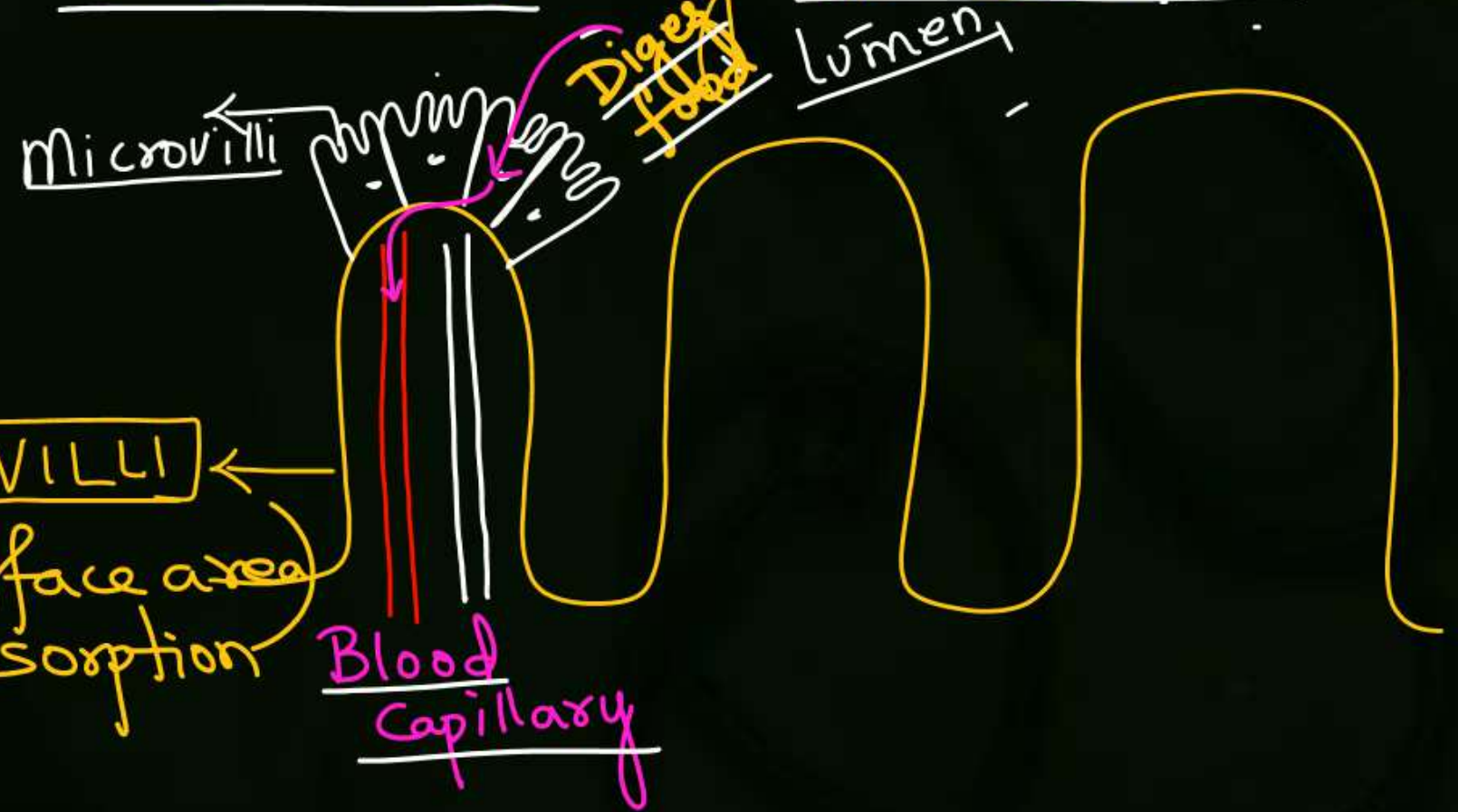


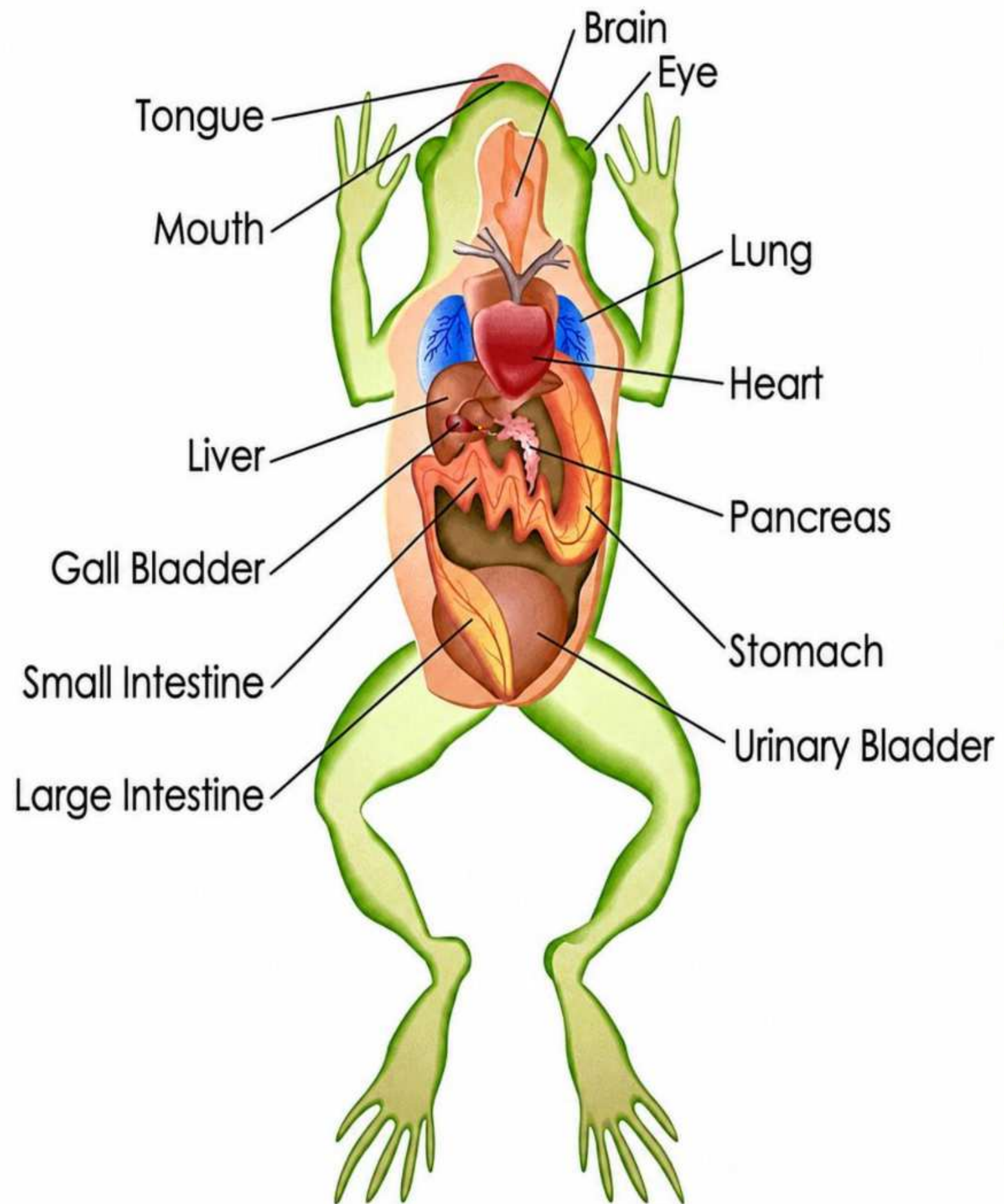
VIII **RECTUM**: receives Undigested Solid waste
 (= Large Intestine)
IX **CLOACA**: Small, median chamber to pass Urine, faecal matter and gametes to exterior





* Digested food is absorbed by numerous finger-like folds in inner wall of intestine, called Villi and microvilli (to ↑ surface area for absorption)





Master NCERT



*Concept se
Question tak!*

7.2 FROGS

Frogs can live both on land and in freshwater and belong to class Amphibia of phylum Chordata. The most common species of frog found in India is *Rana tigrina*.

They do not have constant body temperature i.e., their body temperature varies with the temperature of the environment. Such animals are called cold blooded or poikilotherms. You might have also noticed changes in the colour of the frogs while they are in grasses and on dry land. They have the ability to change the colour to hide them from their enemies (camouflage). This protective coloration is called mimicry. You may also know that frogs are not seen during peak summer and winter. During this period they take shelter in deep burrows to protect them from extreme heat and cold. This is known as summer sleep (aestivation) and winter sleep (hibernation) respectively.

Lec-8

Lec-8

7.2.1 Morphology

Have you ever touched the skin of frog? The skin is smooth and slippery due to the presence of mucus. The skin is always maintained in a moist

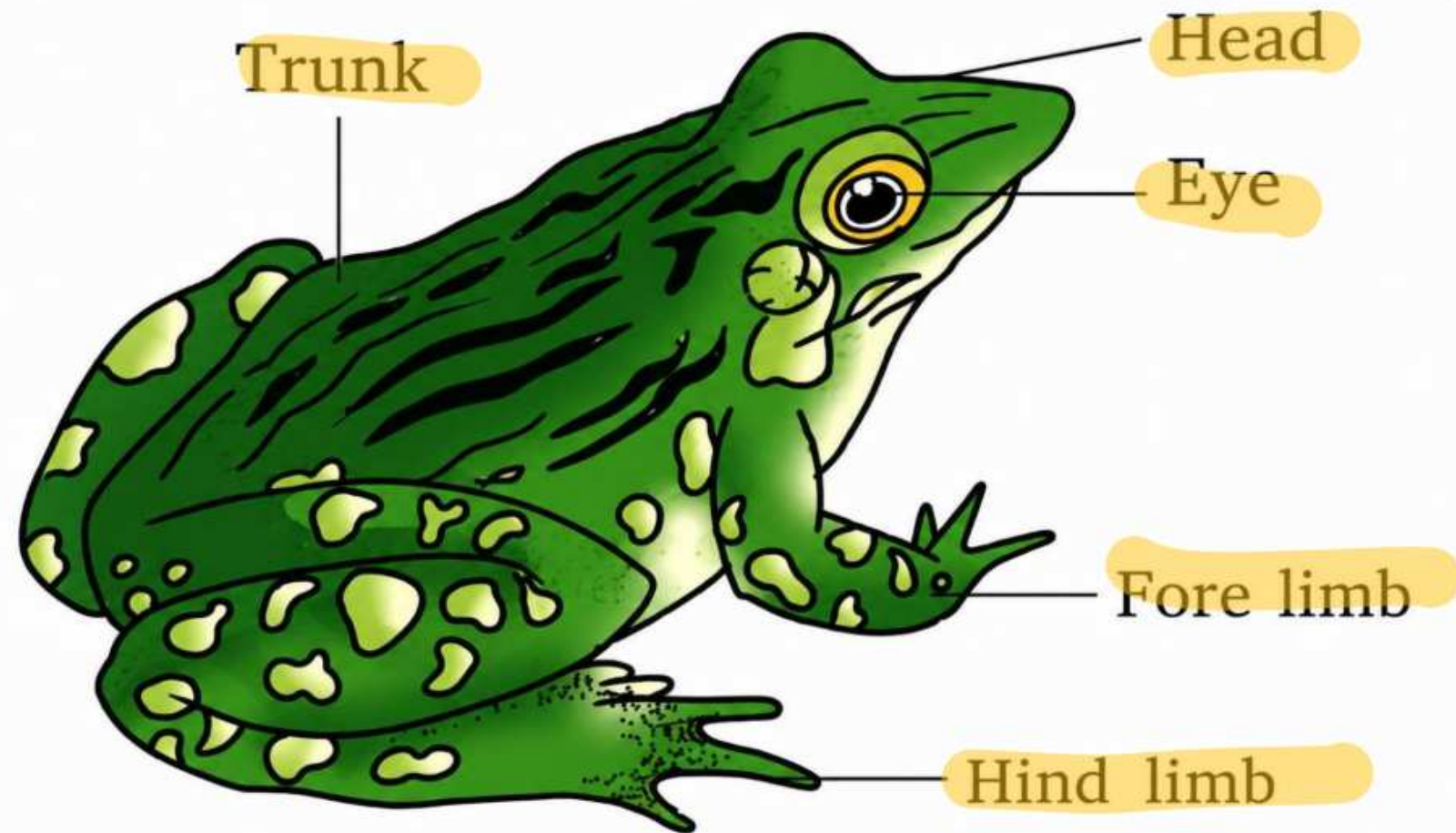


Figure 7.1 External features of frog

condition. The colour of dorsal side of body is generally olive green with dark irregular spots. On the ventral side the skin is uniformly pale yellow. The frog never drinks water but absorb it through the skin.

Body of a frog is divisible into head and trunk (Figure 7.1). A neck and tail are absent. Above the mouth, a pair of nostrils is present. Eyes are bulged and covered by a nictitating membrane that protects them while in water. On either side of eyes a membranous tympanum (ear) receives sound signals. The forelimbs and hind limbs help in

Ver-8

swimming, walking, leaping and burrowing. The hind limbs end in five digits and they are larger and muscular than fore limbs that end in four digits. Feet have webbed digits that help in swimming. Frogs exhibit sexual dimorphism. Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the first digit of the fore limbs which are absent in female frogs.

7.2.2 Anatomy

The body cavity of frogs accommodate different organ systems such as digestive, circulatory, respiratory, nervous, excretory and reproductive systems with well developed structures and functions (Figure 7.2).

The digestive system consists of alimentary canal and digestive glands. The alimentary canal is short because frogs are carnivores and hence the length of intestine is reduced. The mouth opens into the buccal cavity that leads to the oesophagus through pharynx. Oesophagus is a short tube that opens into the stomach which in turn continues as the intestine, rectum and finally opens outside by the cloaca. Liver secretes bile that is stored in the gall bladder. Pancreas, a digestive gland produces pancreatic juice

containing digestive enzymes. Food is captured by the bilobed tongue. Digestion of food takes place by the action of HCl and gastric juices secreted from the walls of the stomach. Partially digested food called chyme is passed from stomach to the first part of the small intestine, the duodenum. The duodenum receives bile from gall bladder and pancreatic juices from the pancreas through a common bile duct. Bile emulsifies fat and pancreatic juices digest carbohydrates and proteins. Final digestion takes place in the intestine. Digested food is absorbed by the numerous finger-like folds in the inner wall of intestine called villi and microvilli. The undigested solid waste moves into the rectum and passes out through cloaca.



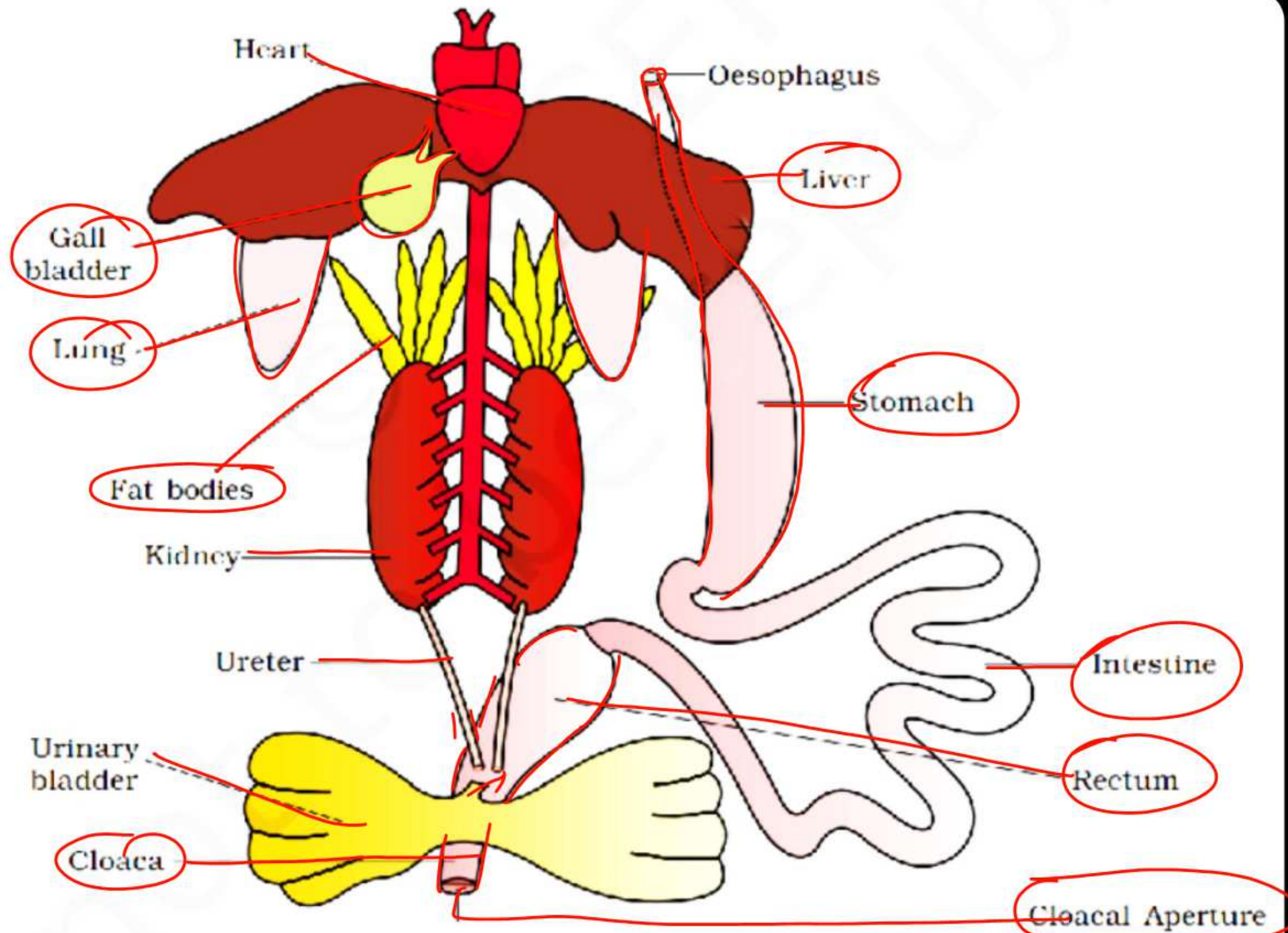


Figure 7.2 Diagrammatic representation of internal organs of frog showing complete digestive system

Frogs respire on land and in the water by two different methods. In water, skin acts as aquatic respiratory organ (cutaneous respiration). Dissolved oxygen in the water is exchanged through the skin by diffusion. On land, the buccal cavity, skin and lungs act as the respiratory organs. The respiration by lungs is called pulmonary respiration. The lungs are a pair of elongated, pink coloured sac-like structures present in the upper part of the trunk region (thorax). Air enters through the nostrils into the buccal cavity and then to lungs. During aestivation and hibernation gaseous exchange takes place through skin.

QUESTION (Mastering NCERT)



The alimentary canal of frog is short because frogs are

- A** herbivores
- B** carnivores
- C** omnivores
- D** heterotrophs

QUESTION (Mastering NCERT)



During aestivation and hibernation of frog gaseous exchange takes place through the.

- ☒ **A** skin
- ☐ **B** nose
- ☐ **C** lungs
- ☐ **D** scales

QUESTION (Mastering NCERT)



Male frog can be distinguished from female frog due to the presence of

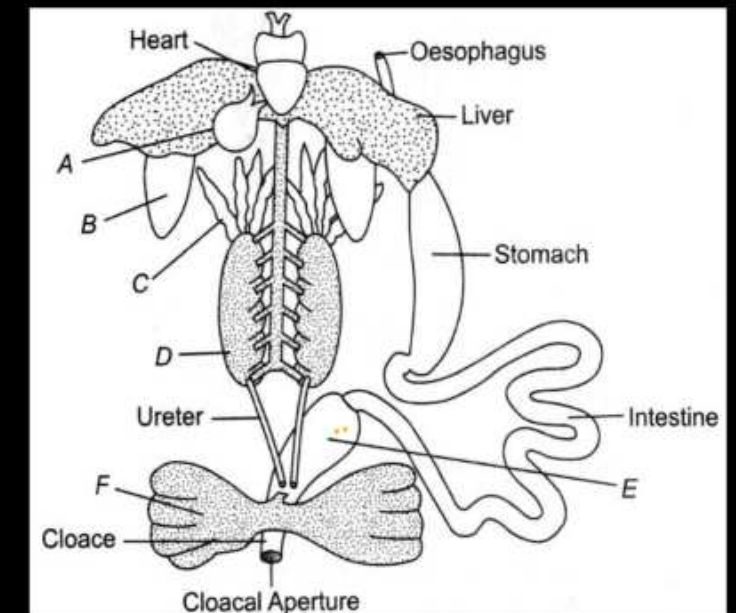
- A** vocal sacs and copulatory pad on the first digit of the forelimb
- B** a neck and tail is absent *Both*
- C** the hind limb ends in the five digits *Both*
- D** eyes are bulged and are covered by the nictitating membrane *Both*

QUESTION (Mastering NCERT)



Given below the diagram of internal organs of frog and identify A to F.

- A** A-Gall bladder, B-Lungs, C-Testis, D-Kidney, E-Urethra, F-Urinary bladder
- B** A-Gall bladder, B-Lungs, C-Fat bodies, D-Kidney, E-Rectum, F-Urinary bladder
- C** A-Gall bladder, B-lungs, C-Ovary, D-Kidney, E-Ileum, F-Urinary bladder
- D** A-Gall bladder, B-Lungs, C-Fat bodies, D-Kidney, E-Colon, F-Urinary bladder





Weekly Test Dose : CPR Given



CPR



SAVES LIVES

H.W
19 Ques
20 min timer

QUESTION (Mastering NCERT)



Assertion (A): Male frogs can be distinguished by the presence of sound producing vocal sacs and a copulatory pad on the first digit of the forelimbs.

Reason (R): These structures help in camouflage and defence against predators.

- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is not the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



Match the List-I with List-II.

Choose the correct answer from the options given below:

- A** A-I, B-II, C-III, D-IV
- B** A-II, B-III, C-IV, D-II
- C** A-III, B-I, C-II, D-IV
- D** A-II, B-IV, C-I, D-III

	List-I		List-II
(A)	Axon	(I)	Receives incoming signals
(B)	Dendrite	(II)	Contains the nucleus
(C)	Cell body	(III)	Transmits signals away from cell body
(D)	Neuroglia	(IV)	Provide support and protection to neurons

QUESTION (Mastering NCERT)



Statement I: Frog have constant body temperature i.e., their body temperature varies with the temperature of the environment. Such animals are called cold blooded or poikilotherms.

Statement II: Male frogs can be distinguished by the presence of a copulatory pad on the last digit of the fore limbs which are absent in female frogs.

- A** Statement I is correct, and Statement II is incorrect.
- B** Statement I is incorrect, and Statement II is correct.
- C** Both Statement I and Statement II are correct.
- D** Both Statement I and Statement II are incorrect.

QUESTION (Mastering NCERT)



Assertion (A): A frog placed in oxygen-deficient water will likely die even if its lungs are fully functional.

Reason (R): While lungs are present, the primary method of oxygen uptake in water for a frog is through its skin, which requires dissolved oxygen.

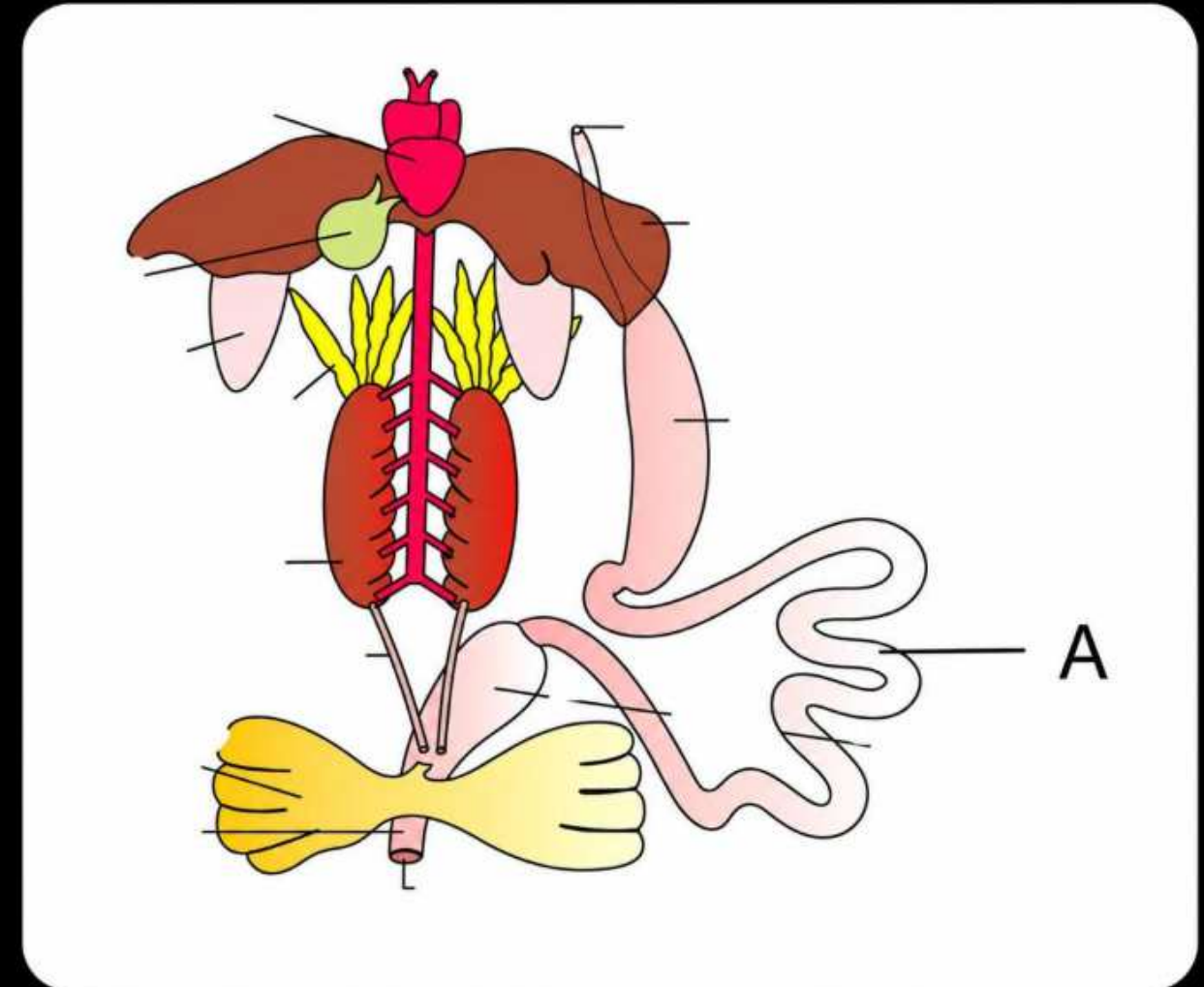
- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is NOT the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



The convoluted structure labeled 'A' is crucial for nutrient assimilation. What is the most significant adaptation present in structure 'A' that maximizes its efficiency in this role?

- A** Presence of a large muscular wall for peristalsis
- B** Extensive folding and villi to increase surface area
- C** Production of strong digestive acids
- D** Storage of undigested food material



QUESTION (Mastering NCERT)



Consider the following statements regarding the frog's circulatory system:

- I. The frog possesses both a well-developed vascular system and a lymphatic system.
- II. The lymphatic system in frogs is primarily responsible for blood transport.
- III. The heart of a frog is a four-chambered structure with two atria and two ventricles.
- IV. The sinus venosus receives blood from the major veins before it enters the left atrium.

Which of the above statements are correct?

- A** I only
- B** I and II
- C** I and IV
- D** I, II, and III

QUESTION (Mastering NCERT)



Statement 1: The blood of a frog is composed of plasma and cells, and the RBCs are nucleated.

Statement 2: The lymph in frogs is similar to blood but lacks RBCs and few proteins.
Which of the following is true?

- A** Statement 1 is true, but Statement 2 is false.
- B** Statement 1 is false, but Statement 2 is true.
- C** Both Statement 1 and Statement 2 are true.
- D** Both Statement 1 and Statement 2 are false.

QUESTION (Mastering NCERT)



Consider the following statements regarding the frog's digestive system:

- I. The alimentary canal is relatively short due to the frog's carnivorous diet.
- II. The mouth opens into the buccal cavity, which then leads to the pharynx.
- III. Bile is produced by the pancreas and stored in the gall bladder.
- IV. The duodenum receives bile and pancreatic juices, both essential for digestion.

Which of the above statements are incorrect?

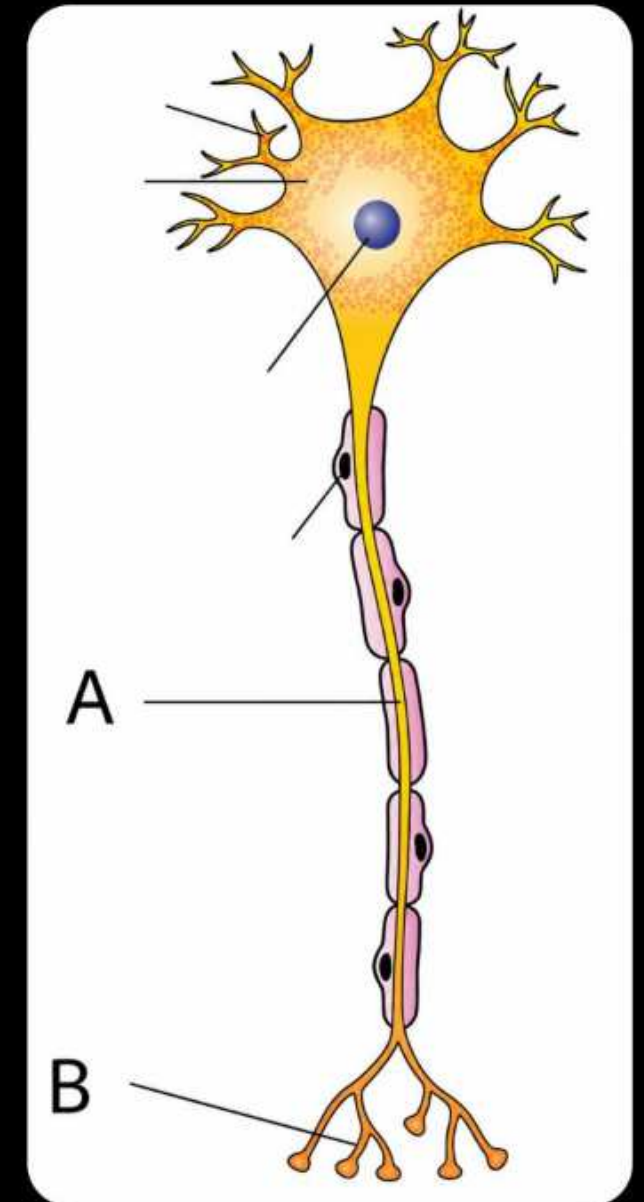
- A** I and II only
- B** III only
- C** I, II, and IV only
- D** I, III, and IV only

QUESTION (Mastering NCERT)



Observe the provided diagram of neural tissue. Identify the part labelled with its correct primary function.

- A** Part B is 'Axon' is the main receptive part of the neuron, collecting signals.
- B** Part A is 'Axon' is the long slender projection that conducts electrical impulses away from the cell body.
- C** Part B is 'Axon terminal' is the central part of the neuron containing the nucleus.
- D** Part A is 'Myelin sheath' is a supporting cell surrounding the neuron.



QUESTION (Mastering NCERT)



Statement 1: The undigested food in a frog's digestive system moves from the rectum into the cloaca and then passes out of the body.

Statement 2: The oesophagus is a long tube that connects the pharynx to the stomach in a frog.

- A** Both Statement 1 and Statement 2 are correct.
- B** Statement 1 is correct but Statement 2 is incorrect.
- C** Statement 1 is incorrect but Statement 2 is correct.
- D** Both Statement 1 and Statement 2 are incorrect.

QUESTION (Mastering NCERT)



Assertion (A): The frog's digestive system is considered "complete."

Reason (R): It possesses a mouth and an anus (cloacal aperture) for the ingestion and egestion of food, respectively.

- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is NOT the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



Which of the following statements about nervous tissue are incorrect?

- I. Neural tissue primarily controls the body's involuntary actions.
- II. Neurons are excitable cells responsible for generating and transmitting electrical disturbances.
- III. Neuroglial cells constitute the major portion of the nervous system's volume.
- IV. Neuroglial cells provide protection and support to neurons.

- A** I, II and III only
- B** II, III and IV only
- C** I, II and IV only
- D** I only

QUESTION (Mastering NCERT)



Assertion (A): Neural tissue exhibits excitability, allowing it to respond to stimuli.

Reason (R): Both neurons and neuroglial cells contribute equally to the generation and transmission of electrical impulses in the nervous system.

- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is NOT the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



How many of the following statements are incorrect about the morphological characteristics of a frog?

- I. The skin is always moist and slippery due to the presence of mucus.
- II. The colour of the dorsal side of the body is generally uniformly pale yellow.
- III. The ventral side of the skin is generally olive green with dark regular spots.
- IV. A pair of nostrils is present above the mouth.

- A** two
- B** four
- C** three
- D** one

QUESTION (Mastering NCERT)



Assertion (A): The skin of a frog is generally olive green with dark irregular spots on the ventral side and uniformly pale yellow on the dorsal side.

Reason (R): Frogs show mimicry, helping them hide from their enemies.

- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is not the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



Match the List-I with List-II.

Choose the correct answer from the options given below:

- A** A-II, B-I, C-IV, D-III
- B** A-IV, B-III, C-I, D-II
- C** A-III, B-I, C-II, D-IV
- D** A-III, B-IV, C-II, D-I

	List-I		List-II
(A)	Poikilotherms	(I)	Male frog's distinguishing characteristic during breeding season
(B)	Mimicry	(II)	Process of taking shelter in deep burrows during summer
(C)	Aestivation	(III)	Animals whose body temperature varies with the environment
(D)	Vocal sacs	(IV)	Protective coloration that helps in hiding from enemies

QUESTION (Mastering NCERT)



Statement 1: During aestivation, a frog's primary mode of gaseous exchange is through its lungs.

Statement 2: The buccal cavity acts as a primary respiratory organ for frogs when they are fully submerged in water.

- A** Both Statement 1 and Statement 2 are true.
- B** Both Statement 1 and Statement 2 are false.
- C** Statement 1 is true, but Statement 2 is false.
- D** Statement 1 is false, but Statement 2 is true.

QUESTION (Mastering NCERT)



Which of the following statements about frog respiration are correct?

- I. Frogs can respire both on land and in water.
- II. Cutaneous respiration involves the exchange of gases through the skin.
- III. Pulmonary respiration occurs when frogs are submerged in water.
- IV. During hibernation, gaseous exchange primarily occurs through the lungs.

- A** I and II only
- B** I, II, and III only
- C** I, II, and IV only
- D** I, II, III, and IV

QUESTION (Mastering NCERT)



Assertion (A): A frog kept submerged in water for an extended period will primarily rely on cutaneous respiration.

Reason (R): The lungs of a frog are adapted for gaseous exchange only in an aerial environment.

- A** Both A and R are true and R is the correct explanation of A.
- B** Both A and R are true but R is NOT the correct explanation of A.
- C** A is true but R is false.
- D** A is false but R is true.

QUESTION (Mastering NCERT)



Match the List-I with List-II.

Choose the correct answer from the options given below:

A A-II, B-I, C-IV, D-III

B A-IV, B-III, C-I, D-II

C A-III, B-I, C-II, D-IV

D A-III, B-IV, C-I, D-II

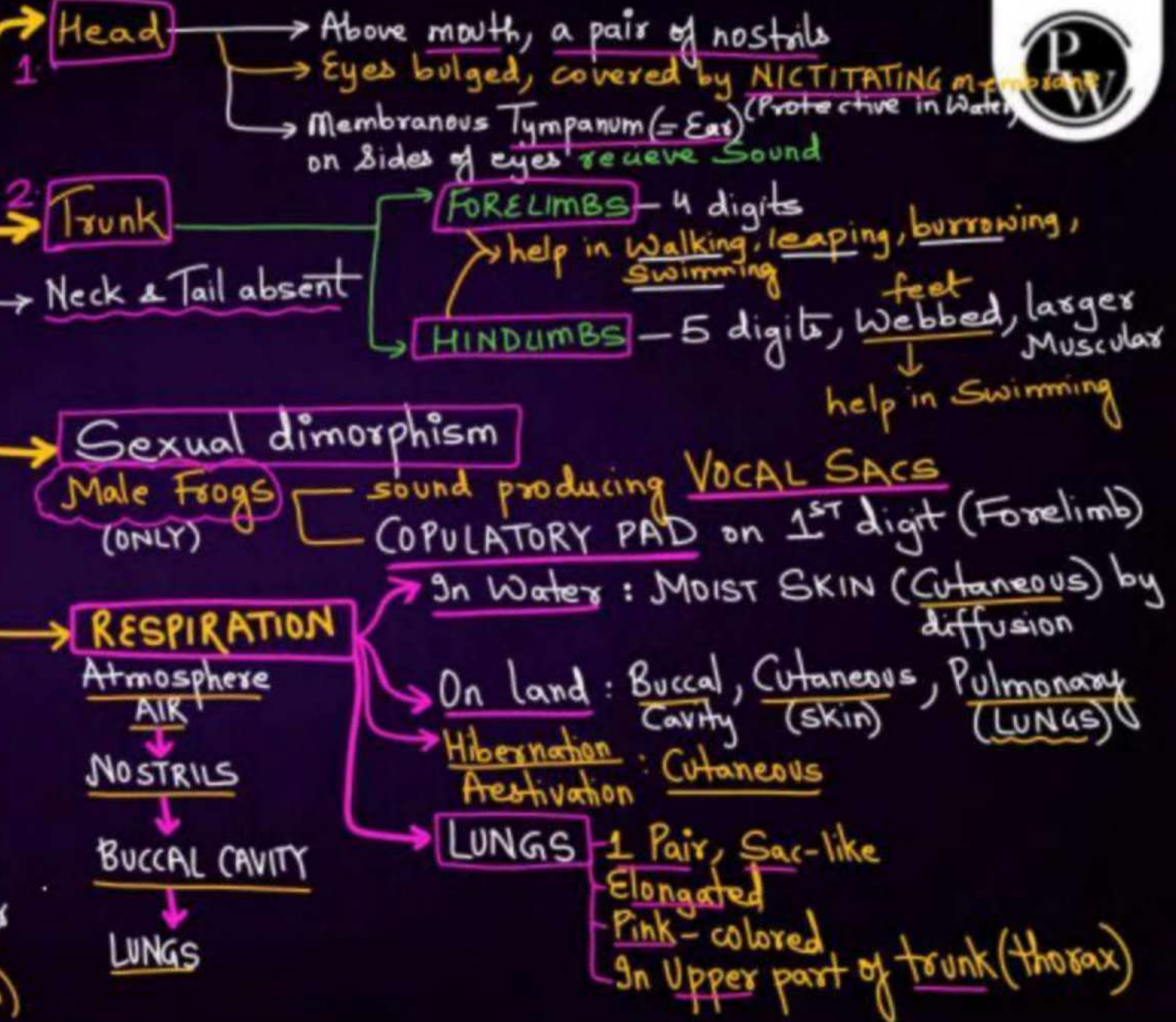
	List-I		List-II
(A)	Sinus Venosus	(I)	Transport blood from the heart to all parts of the body.
(B)	Conus Arteriosus	(II)	Collects blood from different parts of the body.
(C)	Arterial System	(III)	Joins the right atrium, receives blood from major veins.
(D)	Venous System	(IV)	Sac-like opening from the ventricle on the ventral side.



Immunity Booster = Short Notes

FROG (Rana tigrina)

- Phylum - Chordata
- Class - Amphibia
- live on both land and Fresh-water
- Poikilotherm / Cold-blooded (Body temp. NOT Constant)
- CAMOUFLAGE: Change Color to hide from enemies
Protective Coloration = MIMICRY
- Summer sleep = AESTIVATION
- Winter sleep = HIBERNATION
- Skin Smooth, Slippery due to Mucus
- Dorsal Side: Olive green, dark irregular spots
- Ventral Side: Pale yellow
- Never drink Water (Absorb thru Skin)





Immunity Booster
= Short Notes

DIGESTIVE SYSTEM OF FROG

DIGESTIVE GLANDS

1. **LIVER** - Secrete Bile
↓
Emulsification fat

2. **Gall Bladder** - Store Bile

3. **Pancreas** - Pancreatic juice
↓
Digestive enzymes

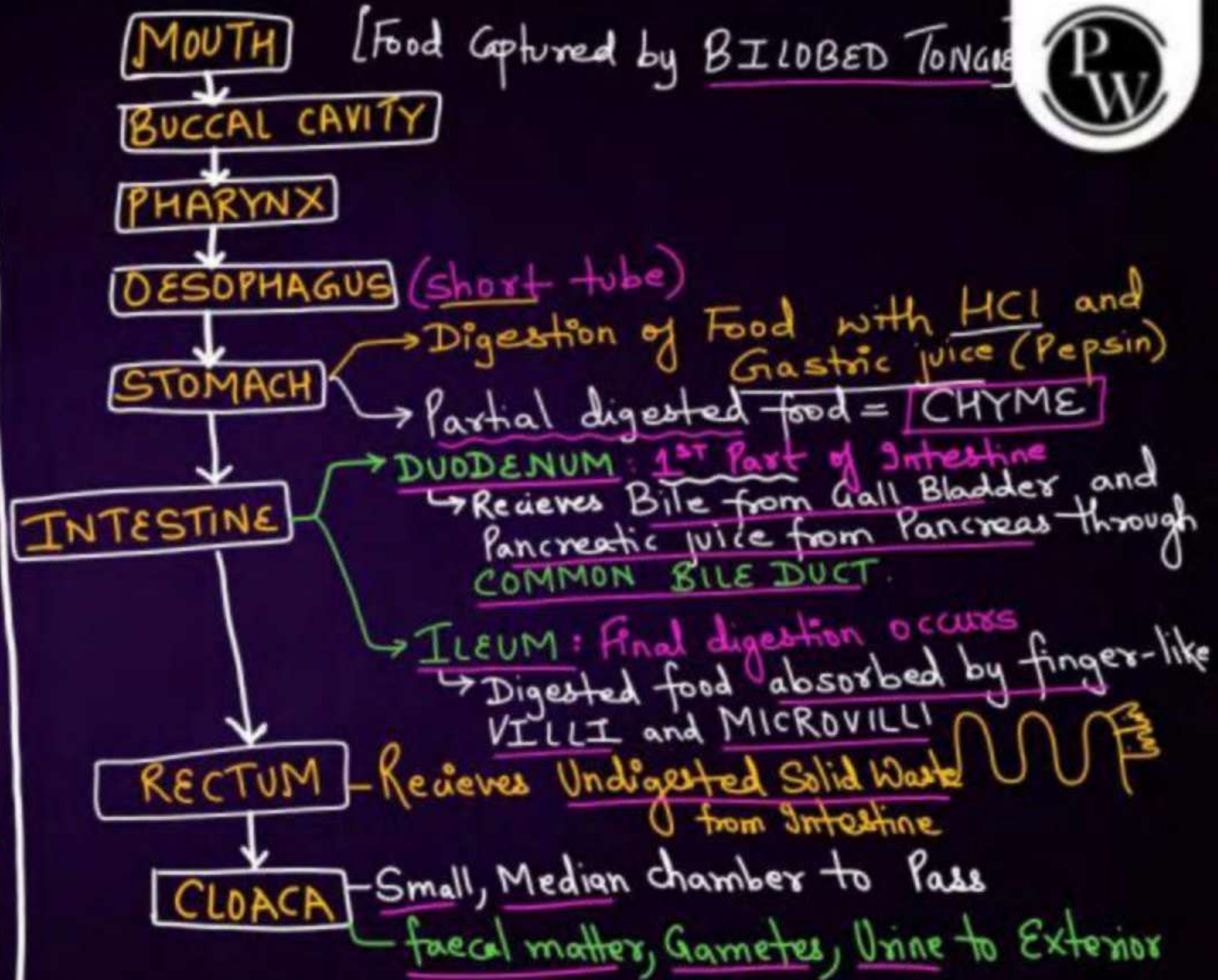
↓
Digest Proteins,
Carbohydrates
↓
Trypsin enzyme

ALIMENTARY CANAL

• SHORT
↓
Foods Carnivore

↓
Intestine length
Reduced

• Mouth to Cloaca





Homework

Module → Today's topics

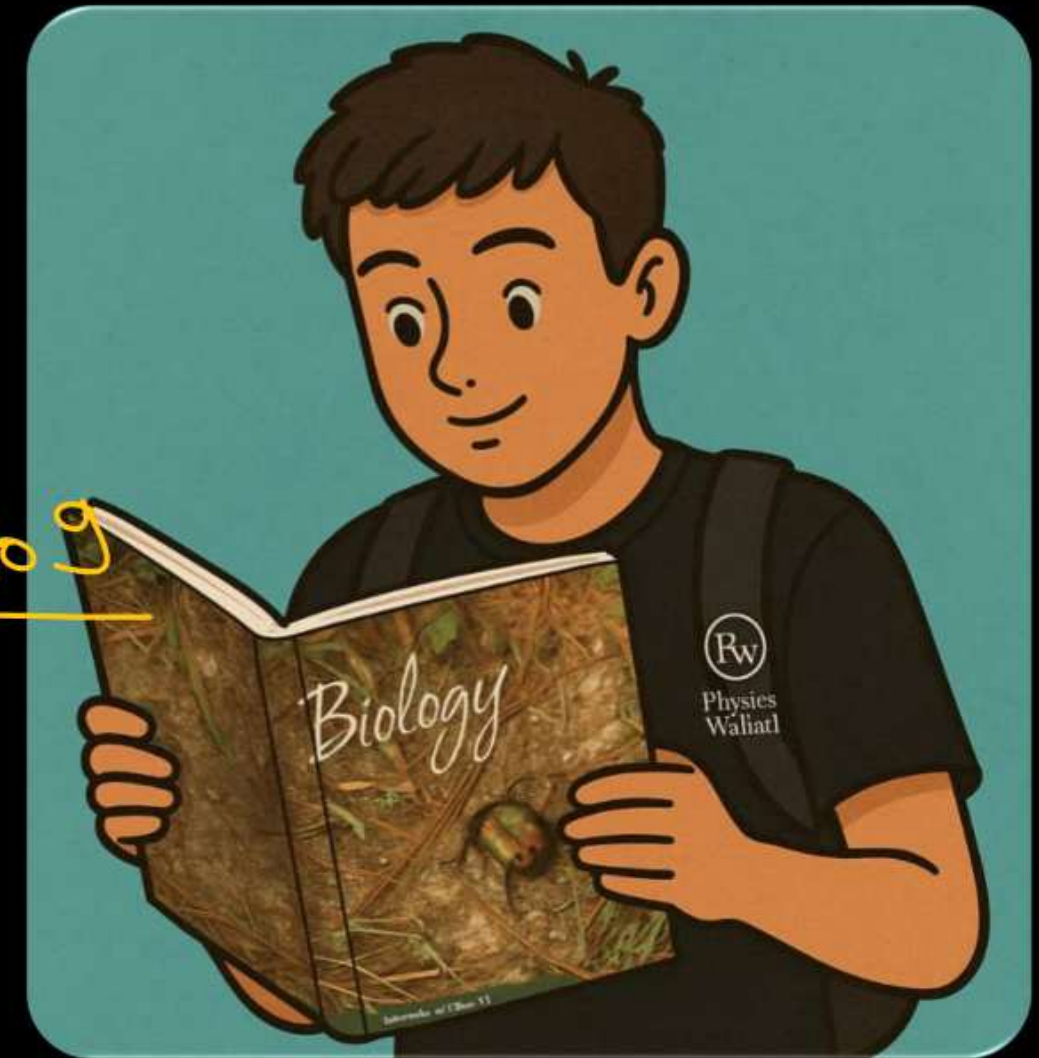
* Revise lec-8

learn lec-9

Read NCERT

Do CPR Ques.

* Sunday: Revise lec-1 to 9



Thank

You

